FIELD PROGRAMME RAHIM – WEEK 4

Predictive Maintenance & Motor Fault Detection

By Friday, you should be able to:

- Understand what predictive maintenance is.
- Understand how Machine Learning can be used to predict equipment failure.
- Work with a real-world predictive maintenance dataset.
- Understand sensor data from industrial machines.
- Prepare a dataset for Machine Learning.
- Identify features and target variables.
- Train a classification model to detect motor faults.
- Evaluate a motor fault detection model.
- Understand the difference between a model that performs well on paper and one that would be useful in the real world.
- Explain the limitations of your model.

Day 1 – Understanding the Motor Problem

Learn

Study:

- What is predictive maintenance?
- Preventive maintenance vs predictive maintenance
- What is equipment failure?
- What is a sensor?
- Why motors generate useful sensor data
- How sensor readings can indicate a potential failure
- What is a fault?
- What is an anomaly?

Think about a motor producing readings such as:

Temperature
Voltage
Current
RPM
Torque

Vibration
Operating Hours

A healthy motor might look like:

Temperature: 60°C
Current: 8.2 A
RPM: 1480
Vibration: 0.15

While a motor developing a problem might look like:

Temperature: 82°C
Current: 11.3 A
RPM: 1290
Vibration: 0.61

The important question is:

Can we use historical sensor data to predict whether a motor is likely to fail?

Practical Work

Program 1 – Explore a Motor Dataset

Find and download a **public predictive-maintenance dataset**.

For example, a dataset containing information about:

- Temperature
- Rotational speed
- Torque
- Tool wear
- Air/process temperature
- Machine failure

One commonly used dataset is the **AI4I 2020 Predictive Maintenance Dataset**.

Load the Dataset

Use Pandas.

Display:

- First 5 rows
- Last 5 rows
- Number of rows
- Number of columns
- Column names
- Data types
- Missing values

Understand the Dataset

For every column, try to understand:

What does this value represent?

For example:

Temperature
→ How hot the machine is.

Rotational Speed
→ How fast the machine is rotating.

Torque
→ Rotational force being produced.

Tool Wear
→ How much the tool has been used.

Important Task

Identify:

Features

Which columns could help us predict a failure?

Target

Which column tells us whether the machine failed?

Write your answers in:

Git Task

Commit and push today's work.

Day 1 - Motor Dataset Exploration

Day 2 – Preparing the Data

Today we learn an extremely important part of Machine Learning:

Data preparation.

A Machine Learning model cannot simply be given a messy dataset and expected to work properly.

Learn

Study:

- Data cleaning
- Missing values
- Duplicate records
- Categorical data
- Numerical data
- Feature selection
- Encoding categorical variables
- Why data preprocessing matters

Practical Work

Using the motor dataset:

Task 1 – Check Missing Values

Find out whether any columns contain missing values.

Task 2 – Check Duplicates

Find duplicate records.

If duplicates exist:

- Count them.

- Decide whether they should be removed.

Task 3 – Check Data Types

Identify which columns are:

Numerical

and which are:

Categorical

Task 4 – Prepare the Dataset

Create:

X

containing the features.

Create:

y

containing the target.

For example:

X = Temperature + Speed + Torque + Tool Wear

y = Machine Failure

Task 5 – Train/Test Split

Split the dataset into:

80% Training
20% Testing

Understand why we do this.

Deliverable

Create:

prepare_motor_data.py

The program should:

1. Load the dataset.
2. Clean the dataset.
3. Select features.
4. Select target.
5. Split training/testing data.

Git Task

Commit and push.

Day 3 – Train the First Motor Fault Model

Now we actually train a model.

Learn

Study:

- Classification
- Decision Trees
- Random Forest
- Training a model
- Making predictions

Practical Work

Start with:

Model 1 – Decision Tree

Train a Decision Tree Classifier using the motor dataset.

Process:

```
Motor Dataset
   ↓
Prepare Data
   ↓
X + y
   ↓
Train/Test Split
   ↓
Decision Tree
   ↓
Train
   ↓
Predict
```

Make Predictions

Use the testing dataset.

Generate:

Actual
Predicted

Example:

Actual: 0
Predicted: 0

Actual: 1
Predicted: 1

Actual: 0
Predicted: 1

Calculate

- Accuracy
- Precision
- Recall
- F1-score

Think About This

If the model predicts:

FAULT

but the motor is actually healthy:

That's a **false positive**.

If the model predicts:

HEALTHY

but the motor actually fails:

That's a **false negative**.

Ask him:

Which one could be more dangerous in a real industrial environment?

Git Task

Commit and push.

Day 3 - First Motor Fault Model

Day 4 – Improving & Understanding the Model

Today we don't just train one model and celebrate the accuracy.

We investigate it.

Learn

Study:

- Random Forest
- Feature importance
- Confusion matrix
- Overfitting
- Model comparison

Practical Work

Model 1

Decision Tree

Model 2

Random Forest

Train both using the same dataset.

Compare

Create a table:

Model	Accuracy	Precision	Recall	F1

Decision Tree	XX%	XX%	XX%	XX%
Random Forest	XX%	XX%	XX%	XX%

Feature Importance

Use the Random Forest model to identify which features were most important.

For example, he might discover:

The exact results will depend on the dataset.

Important Discussion

Ask:

"Does feature importance prove that this sensor causes the motor to fail?"

The answer should be **no**.

Feature importance tells us that the model found the feature useful for making predictions. It does **not** prove causation.

This is a very important concept for his final-year project.

Deliverable

Create:

motor_model_comparison.py

and produce:

model_comparison.txt

explaining which model performed better and why.

Day 5 – Mini Project

Motor Fault Detection System – Version 1

Now he combines everything from Weeks 2–4.

Objective

Create a Python program that takes motor sensor information and predicts whether the motor is likely to experience a failure.

Program Requirements

The program should:

1. Load the dataset

Use Pandas.

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2. Prepare the data

Clean and prepare the dataset.

---

3. Train the model

Use the better-performing model from Day 4.

For example:

```
Random Forest
```

---

4. Evaluate the model

Display:

Accuracy
Precision
Recall
F1-score
Confusion Matrix

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5. Predict a New Motor

Allow the user to enter sensor values.

For example:

```
Enter Motor Information

Temperature: 75
Rotational Speed: 1400
Torque: 50
Tool Wear: 180
```

The system should output:

```
Prediction:

⚠️ POSSIBLE MOTOR FAILURE
```

or:

```
Prediction:

MOTOR OPERATING NORMALLY
```

Important

The prediction should be based on the **trained ML model**, not simply:

```
if temperature > 70:
    print("Failure")
```

This is the key difference between the Week 2 threshold-based system and the Week 4 ML system.

Documentation

Create:

README.md

Include:

1. Problem

What problem are you trying to solve?

2. Dataset

Where did the data come from?

3. Features

What information did you give the model?

4. Target

What are you trying to predict?

5. Models

Which models did you test?

6. Results

How did they perform?

7. Best Model

Which model did you choose?

8. Limitations

What could make your model inaccurate?

9. Future Improvements

What would you improve?